

Certificates and reductions

Exercise sheet 8

Exercise 8.1. Prove that the following problems are in NP:

- (a) $\{\langle G, s, t, k \rangle \mid \text{graph } G \text{ contains a simple path from } s \text{ to } t \text{ visiting at least } k \text{ vertices}\}$;
- (b) $\text{SUBSET-SUM} = \{\langle a_1, \dots, a_m, t \text{ (encoded in binary)} \rangle \mid \exists I \subseteq \{1, \dots, m\} : \sum_{i \in I} a_i = t\}$;
- (c) $3\text{COLOUR} = \{G \mid \text{graph } G \text{ admits a proper colouring using at most three colours}\}$.

Exercise 8.2. Suppose $L_1, L_2 \in \text{NP}$. Is it true that $L_1 \cup L_2 \in \text{NP}$? Is it true that $L_1 \cap L_2 \in \text{NP}$?

Exercise 8.3. Recall that $L^* = \{w_1 \cdots w_r \mid r \geq 0, w_i \in L\}$. Suppose $L \in \text{NP}$. Is it true that $L^* \in \text{NP}$?

Exercise 8.4. Let $A \leq_p B$ and $B \leq_p C$.

- (a) Prove that $A \leq_p C$.
- (b) If $C \in \text{P}$, what can we say about A ?
- (c) If A is NP-complete, what can we say about C ?

Exercise 8.5. * Suppose that $\text{P} = \text{NP}$. Prove that there is a polynomial-time algorithm that *finds* a satisfying assignment for a given Boolean formula if one exists, and otherwise reports that none exists.

A *nondeterministic* Turing machine M may have more than one transition for a state and observed symbol: for example, both $(q, a) \mapsto (q', b, L)$ and $(q, a) \mapsto (q'', c, R)$ may be possible. Thus M can ‘choose’ which transition to execute. Such a machine *runs in polynomial time* if there is a polynomial p such that *every possible* computation of M on input w halts within $p(|w|)$ steps. We say that M *accepts* w if *some* computation on w halts in the accepting state; it *rejects* w if all computations reject.

Exercise 8.6. * Prove that $L \in \text{NP}$ if and only if it is decided by a polynomial-time nondeterministic Turing machine.

Exercise 8.7. * Let NEXP be the class of languages decided by exponential-time nondeterministic Turing machines. Prove that if $\text{P} = \text{NP}$ then $\text{EXP} = \text{NEXP}$.